

Head Office & Branch Manager Inventory Architecture

Module: Brand Owner / Head Office → Inventory • **Version:** 2.0 • **Scope:** Multi-Branch Stock, Service BOM, Transfers, Consumption & Procurement

Executive Summary & Architectural Context: The Digiflex Salon SaaS platform maintains strict separation of concerns between the **Brand Owner Inventory Desk** and the separate **Inventory / Procurement Operations Panel**. While day-to-day store clerks perform physical receiving and stockroom picking in their operational desk, the Brand Owner interface serves as the centralized governance cockpit—overseeing multi-branch stock valuation, Service Recipe BOM compliance, Maker-Checker PO authorizations, inter-branch balancing transfers, chairside consumption variance leakage, and physical stocktake adjustments.

1. PRD Traceability Matrix

PRD REQUIREMENT ID	FEATURE NAME	PRODUCT REQUIREMENT DESCRIPTION	IMPLEMENTATION DETAILS IN CODE
SALO-PRD-INV-01	Inventory Dashboard & Multi-Branch Comparison	10 centralized KPI cards (Total Stock ₹28.5L, Low Stock, Expired, POs, Transfers, Wastage), Donut Health visualization, and multi-branch comparison table with sorting.	Implemented in <code>InventoryOverviewTab.tsx</code> with reactive filter controls, health distribution bars, and branch comparison ranking.
SALO-PRD-INV-02	Master Products & Consumables Catalogue	Unified catalogue supporting Retail Products and Professional-use consumables with UOM, FIFO cost, selling MRP, reorder limits, and detailed batch profile drawer.	Implemented in <code>ProductsConsumablesTab.tsx</code> with sub-tabs for Retail vs. Consumables and comprehensive product dossier modal.
SALO-PRD-INV-03	Service Recipe / BOM Specifications	Service recipe management mapping exact consumable quantities (ml/gm/unit) to salon services with versioning (e.g. v2.2) and material cost estimates.	Implemented in <code>ProductsConsumablesTab.tsx</code> (Service BOM tab) with recipe inspection, duplication, and historical version changelog.
SALO-PRD-INV-04	Running Stock Ledger & Movement Timeline	Real-time multi-branch stock ledger with available/reserved/damaged counts and chronological movement log with running balance.	Implemented in <code>StockTab.tsx</code> with positive/negative quantity delta badges (+24 / -1) and reference document tracking.
SALO-PRD-INV-05	Batch & Expiry Risk Monitoring	Pre-expiry mitigation system with 7/15/30/60/90 days filters and days-to-expiry gauges to prevent chemical spoilage.	Implemented in <code>StockTab.tsx</code> (Batch & Expiry tab) with quick quarantine and transfer suggestions.
SALO-PRD-INV-06	Procurement & Purchase Order Governance	Maker-checker PO review, receiving progress bars (ordered vs received), vendor directory, GRN inward logs, and purchase returns.	Implemented in <code>ProcurementTab.tsx</code> with PO Dossier modal and cross-links to Finance Panel for accounting settlement.
SALO-PRD-INV-07	Inter-Branch Stock Transfers	Multi-branch stock balancing with 5-stage chain-of-custody progress tracker (Requested → Approved → Dispatched → In Transit → Received).	Implemented in <code>TransfersTab.tsx</code> with source/destination branch inventory validation and HO authorization modal.
SALO-PRD-INV-08	Consumption & Recipe Variance Lifecycle	Full PRD lifecycle: Recipe suggested → Actual confirmed → Stock reduced → Variance flagged → Wastage approved.	Implemented in <code>ConsumptionWastageTab.tsx</code> with stylist leakage detection and loss authorization workflows.

PRD REQUIREMENT ID	FEATURE NAME	PRODUCT REQUIREMENT DESCRIPTION	IMPLEMENTATION DETAILS IN CODE
SALO-PRD-INV-09	Physical Stocktake & Reconciliation	Monthly cycle count verification comparing system book stock against physical shelf counts with discrepancy approval.	Implemented in <code>StocktakeTab.tsx</code> with line-item discrepancy audits and ledger adjustment posting.
SALO-PRD-INV-10	Inventory Reports, Alerts & Audit Trail	6 statutory report categories, multi-format export (CSV, Excel, PDF), 4-severity inventory risk alerts, and immutable audit journal.	Implemented in <code>InventoryReportsTab.tsx</code> with interactive preview canvas and statutory compliance logs.

2. Business Rationale & Why We Implemented Everything

1. Why Centralized Stock Valuation is Critical for Brand Owners

In multi-branch salon networks, inventory represents the second largest current asset after cash. The **FIFO Valuation Engine** computes stock worth across all 5 branches (₹28,50,000), giving leadership immediate visibility into capital locked in backwash chemicals and retail displays without waiting for end-of-month accounting books.

2. Why Service Recipe / BOM (Bill of Materials) is Enforced

Without standardized recipes, colour mixing and treatment applications vary wildly by stylist. By enforcing **Service BOMs (e.g. 50ml Majirel + 75ml Developer)**, the POS automatically calculates expected stock reduction upon service completion. When a stylist dispenses 75ml instead of 60ml (+25% variance), the system flags the leakage instantly, enabling targeted training rather than absorbing unbilled product loss.

3. Why Inter-Branch Stock Transfers Prevent Dead Stock

Slow-moving inventory in smaller branches (e.g., Ujjain or Bhopal) often expires before utilization. The **Pre-Expiry Risk Matrix** alerts Head Office 30 days prior, allowing one-click authorization of an **Inter-Branch Transfer** to high-volume locations like Indore Central, ensuring 100% stock utilization and zero chemical write-off.

4. Why Maker-Checker Purchase Order Approval is Mandatory

To eliminate unauthorized store purchases, storekeepers submit PO requisitions that require **Head Office Brand Owner Sign-Off** before vendor dispatch. Once delivered, Goods Receipt Notes (GRN) verify physical counts before sending summary notices to the separate Finance Panel for bank disbursement.

3. UI Architecture & Page Flow

- **Master Layout (InventoryMasterPage.tsx):** Hosts 8 sub-navigation tabs, breadcrumb context, and reactive search param synchronizer.
- **Overview Tab (InventoryOverviewTab.tsx):** 10 KPI summary cards, interactive shelf pacing distribution bar, and branch comparison table with multi-column sorting.
- **Products & Consumables (ProductsConsumablesTab.tsx):** Retail catalogue, professional consumables, Product Dossier modal with batch inventory, and Service Recipe BOM interface.
- **Stock Tab (StockTab.tsx):** Central stock ledger, running movement journal with delta badges, batch expiration monitor, and stock ageing analysis (0–30d to 180+d).
- **Procurement Tab (ProcurementTab.tsx):** PO register, Vendor directory, GRN inward logs, Purchase Returns, and Finance Panel integration hooks.
- **Transfers Tab (TransfersTab.tsx):** Inter-branch stock transfer ledger with 5-stage chain-of-custody progress tracker.
- **Consumption & Wastage (ConsumptionWastageTab.tsx):** Backwash consumption ledger, Recipe vs. Actual variance analytics, Wastage approvals, and Retail returns.
- **Stocktake Tab (StocktakeTab.tsx):** Physical cycle count ledger with line-by-line discrepancy reconciliation drawer.
- **Reports & Audit (InventoryReportsTab.tsx):** 6 report categories, multi-format export (CSV/Excel/PDF), 4-severity risk alerts, and immutable audit journal.